



A Phase 2, Randomized, Clinical Trial of Inhaled Nitric Oxide for Acute Severe Right Heart Failure With Pulmonary Hypertension (PHiNO Study)

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Background: Acute right heart failure (RHF) is life threatening in patients with pulmonary hypertension (PH). This study investigated the efficacy and safety of inhaled nitric oxide (iNO; INOflo® for inhalation 800ppm), a rapid pulmonary vasodilator, in patients with acute severe RHF due to PH.

Methods and Results: In this Phase 2 randomized controlled trial, 30 patients with acute severe RHF due to PH (pulmonary arterial hypertension or chronic thromboembolic PH) were randomized 1:1 to either an iNO or control group (which did not receive iNO). The primary endpoint was the change in pulmonary vascular resistance (PVR) from baseline to 30 min after either iNO initiation or patient registration and assignment (control group), which differed significantly between the iNO and control groups (mean [±SD] -2.41 ± 2.47 vs. 0.8 ± 1.03 Wood units, respectively; between group difference -3.21 [95% confidence interval $-4.633, -1.785$] Wood units). Serum B-type natriuretic peptide (BNP) levels and inferior vena cava diameter (secondary endpoints) decreased significantly in the iNO group over the 7-day study period. No serious adverse events, including methemoglobinemia, were observed.

Conclusions: iNO significantly reduced PVR in patients with acute severe RHF due to PH, without serious adverse events. Serum BNP levels and inferior vena cava diameter improved in the iNO group. These findings suggest that iNO is a promising acute treatment option for RHF due to PH.

Key Words: Chronic thromboembolic pulmonary hypertension; Nitric oxide; Pulmonary arterial hypertension; Pulmonary hypertension; Right heart failure

Acute right heart failure (RHF) is characterized by a sudden loss of right ventricular function, leading to systemic hypoperfusion.^{1,2} Pulmonary hypertension (PH) is a cause of RHF,³ and patients with acute severe RHF due to PH have a poor prognosis if they remain untreated. The recommended treatments for RHF with PH are to: (1) address triggers and provide supportive therapy; (2) optimize fluid status using diuretics or hemofiltration; (3) reduce right ventricular afterload; (4) optimize cardiac output; and (5) optimize blood pressure.⁴ Among these treatments, reducing right ventricular afterload is a specific treatment for patients with RHF due to PH,⁵ and pulmonary vasodilators may be an option for such patients.

Drugs used for pulmonary arterial hypertension (PAH)

and chronic thromboembolic PH (CTEPH)^{3,6-8} target three pathways and dilate the pulmonary artery; they may also improve RHF caused by PH.⁴ However, currently approved PH drugs have limitations in the treatment of the acute phase of RHF. Specifically, orally administered PH drugs require a considerable period of time of optimal efficacy to be achieved.³ In addition, intravenous and subcutaneous drugs, such as epoprostenol, require dose escalation, thereby extending the time required to achieve therapeutic effects. Although inhaled medications offer a relatively rapid onset of action, they produce transient maximum effects, necessitating frequent administration and complicating the maintenance of consistent and therapeutic drug concentrations. Moreover, these agents pose a risk of systemic hypo-

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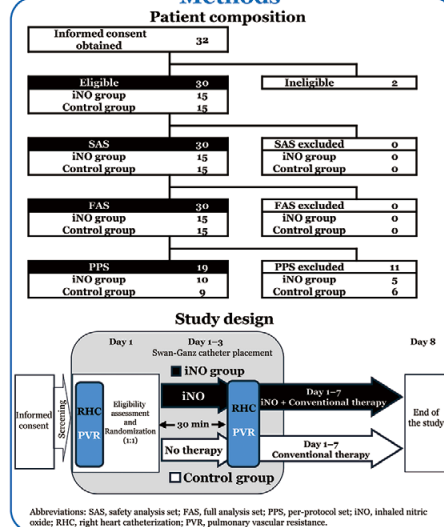
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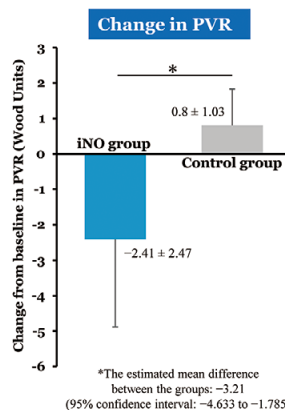
Background: This study aimed to investigate the efficacy and safety of inhaled nitric oxide (iNO) (INOflor[®] for inhalation 800 ppm) in acute severe right heart failure due to pulmonary hypertension.

Methods

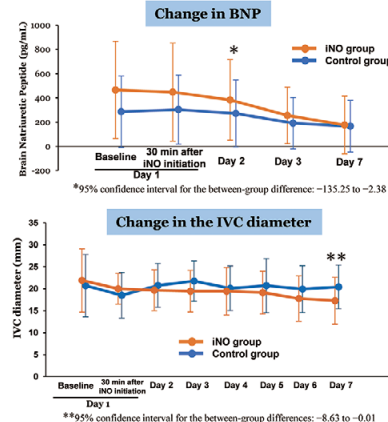


Results

Primary endpoint (FAS)



Secondary endpoints (FAS)



Safety (SAS): No serious adverse events, including methemoglobinemia, were observed.

Abbreviations: SAS, safety analysis set; FAS, full analysis set; iNO, inhaled nitric oxide; PVR, pulmonary vascular resistance; BNP, brain natriuretic peptide; IVC, inferior vena cava.

Conclusions: iNO significantly reduced pulmonary vascular resistance in patients with acute severe right heart failure due to pulmonary hypertension, with no serious adverse events. Serum brain natriuretic peptide levels and inferior vena cava diameter improved in the iNO group.

Central Figure

tension due to their systemic vasodilatory effects.⁹ In the acute treatment of patients with new-onset PH, it typically takes several days or longer after a patient is admitted to hospital before the administration of currently approved PH medications is started because initial evaluations, such as identifying the cause of PH and determining its severity, are needed before treatment can be initiated. However, before approved PH drugs are initiated and their effects are observed, there is no immediate and safe treatment available to reduce the afterload on the right ventricle. Consequently, patients with severe PH can exhibit rapid deterioration, and even die, during the acute phase, which presents a clinical challenge and highlights a gap in clinical practice. Therefore, for patients with RHF due to PH with severe and unstable acute hemodynamics, there is an urgent need for pulmonary vasodilators that act quickly and have better safety profiles to bridge the gap to the currently approved PH medications.

Inhaled nitric oxide (iNO) rapidly dilates the pulmonary artery, reduces right ventricular afterload, and improves hemodynamics in acute right heart dysfunction.¹⁰ Owing to its rapid action and selective pulmonary vasodilator effect with minimal systemic vascular impact,^{9,11} iNO may be an ideal therapeutic option for acute severe RHF with PH¹²⁻¹⁵ and may serve as a safe transition to subsequent PH treatments in acute RHF with PH. In Japan, INOflor[®] (Mallinckrodt Manufacturing LLC, NJ, USA and Air Water Inc., Tokyo, Japan) for inhalation (800 ppm) has been approved for 2 specific indications: hypoxic respiratory failure with persistent PH in neonates¹⁶ since 2008; and PH during the perioperative period of cardiac surgery since 2015. Despite the promising properties of iNO, robust evidence for its use in acute RHF with PH in adults, beyond the

neonatal and perioperative settings, remains limited. Therefore, we conducted a Phase 2 randomized controlled trial to investigate the efficacy and safety of iNO as an acute phase treatment for severe RHF with PH (PHiNO study; Japan Registry of Clinical Trials [jRCT] ID: jRCT 2051220042).

Methods

Study Design

The PHiNO study is a Phase 2 investigator-initiated randomized open-label parallel-group comparison and control trial designed to examine the efficacy and safety of iNO over a 7-day period as an acute treatment after admission of patients with severe RHF due to PH. The study was conducted at the National Cerebral and Cardiovascular Center in Japan. The clinical trial started on November 2, 2022 (the date the first patient consented) and concluded on September 18, 2024 (the date of the last patient observation, including follow-up). The study protocol was published as a protocol paper.¹⁷

This study was approved by the Institutional Review Board of the National Cerebral and Cardiovascular Center (Approval #1203). All procedures complied with the Good Clinical Practice guideline of the International Conference on Harmonization of Technical Requirements for Registration of Pharmaceuticals for Human Use and its revised ministerial ordinances/operational notifications in Japan. This study was conducted in compliance with the Declaration of Helsinki of the World Medical Association (1964) and the Ministerial Ordinance Concerning the Standards for the Conduct of Clinical Trials of Pharmaceuticals. Written informed consent was obtained from all patients prior to their enrollment. If the written informed consent form was

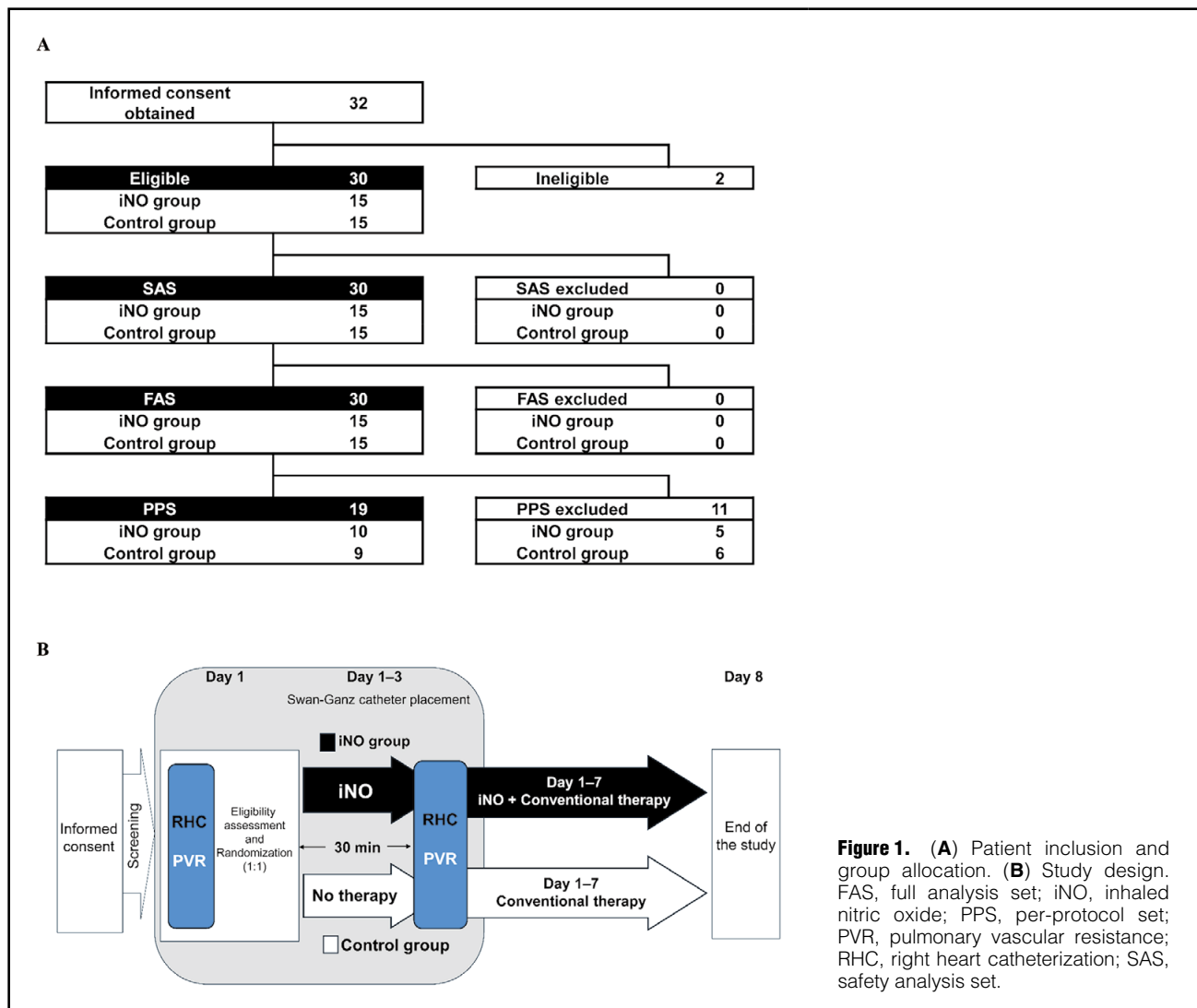


Figure 1. (A) Patient inclusion and group allocation. (B) Study design. FAS, full analysis set; iNO, inhaled nitric oxide; PPS, per-protocol set; PVR, pulmonary vascular resistance; RHC, right heart catheterization; SAS, safety analysis set.

revised, the study physicians explained the procedure again, revised the written informed consent form, and obtained the participant's voluntary consent to continue participation.

Patient Enrollment

Patient enrollment and group allocation are shown in **Figure 1A**. The key inclusion and exclusion criteria and the reasons for ineligibility are listed in **Supplementary Table 1**. **Supplementary Table 2** provides detailed reasons for exclusion from the study. The safety analysis set (SAS) comprised all patients registered in this clinical trial who received the investigational treatment. The full analysis set (FAS) comprised patients enrolled in the trial who received the investigational treatment and for whom data were observed for at least one efficacy endpoint. The FAS was used as the population for the efficacy analysis of the primary and secondary efficacy endpoints. Within the FAS, the per-protocol set (PPS) was defined as patients who complied with the protocol, excluding patients with protocol deviations or violations that could impact efficacy assessment (e.g., patients deviating from the inclusion criteria, subjects breaching the exclusion criteria, and patients contravening

prohibited concomitant medication regulations). As a supplementary analysis to assess whether the extent of deviations or violations from the trial protocol significantly affected the primary analysis results based on the FAS, we performed a primary endpoint analysis using the PPS. Secondary endpoints (safety assessment) were analyzed using the SAS.

To be eligible for inclusion in the study, patients must have been diagnosed with PAH or CTEPH, have undergone right heart catheterization (RHC), and have a mean pulmonary artery pressure (mPAP) ≥ 25 mmHg at rest, pulmonary artery wedge pressure or left ventricular end-diastolic pressure ≤ 15 mmHg, and pulmonary vascular resistance (PVR) > 3 Wood units. All patients with PAH underwent acute vasoreactivity testing (AVT) at the during the first RHC using iNO at a concentration of 20 ppm for 10 min via a facemask. AVT responders were defined as those who exhibited a reduction in mPAP ≥ 10 mmHg to an absolute value of ≤ 40 mmHg with maintained or increased cardiac output, as per the guideline.³ The investigators who participated in this study were not involved in the AVT and were not informed about the AVT results.

We defined 'severe' RHF as World Health Organization

functional class III or higher, with fluid retention (on physical examination or echocardiographic evidence of congestion) and a low cardiac output (cardiac index <2.5 L/min/m² or the need for inotropic agents).

The study design is shown in **Figure 1B**. Patients underwent a series of tests, including blood tests, arterial blood tests, echocardiography, chest radiography, and RHC, to determine their eligibility for the study. Data from computed tomography and perfusion scintigraphy conducted within 365 days before enrollment were analyzed to diagnose PAH and CTEPH. Following RHC, eligible patients were randomized in a 1:1 ratio to either the iNO or control group, using Pocock and Simon's minimization method,¹⁸ with disease (CTEPH/PAH) as a prognostic factor. Patient enrollment and assignment were managed using an electronic data capture system.

Intervention

iNO Group Patients assigned to the iNO group received iNO for 30 min until PVR measurement and continued iNO over a 7-day treatment period (until Day 7 of hospitalization). The study drug, INOflo, was supplied by Mallinckrodt Manufacturing LLC and Air Water, Inc. iNO therapy was initiated at a concentration of 20 ppm; if the blood methemoglobin (MetHb) concentration remained below 2%, the dose was increased by 10 ppm every 5 min, to a maximum of 40 ppm. The administration route for iNO in the study protocol was designed to allow for its delivery via methods that did not require intubation, including nasal high-flow oxygen therapy, nasal continuous positive airway pressure, and face masks, in addition to its use in intubated patients who were managed with ventilators, depending on patient condition. PVR was assessed 30 min after the initiation of iNO inhalation. This was chosen as the primary endpoint due to the rapid onset of action of iNO and the acute nature of RHF. Following completion of PVR evaluation, iNO was continued until Day 7. Conventional therapy, which included inotropic agents, diuretics, vasopressors, and other guideline (2017)-recommended therapies for PH⁴ administered according to a patient's condition, was not allowed until PVR had been measured 30 min after iNO initiation but was allowed thereafter. Patients were provided with a non-intubated oxygen source (e.g., nasal high-flow, nasal continuous positive airway pressure, or face mask) and intubation and extracorporeal membrane oxygenation, if necessary.

During iNO therapy, the MetHb concentration was assessed using blood tests and pulse oximetry (Masimo Rainbow SET) to facilitate non-invasive, continuous, and immediate monitoring. If the MetHb concentration exceeded 2.5% but was less than 4%, the iNO concentration was decreased by 5 ppm and subsequently reduced by an additional 5 ppm every 5 min. If the MetHb concentration was successfully reduced to $\leq 2\%$, the iNO concentration was incrementally increased by 5 ppm every 5 min until a dosage was achieved that maintained a MetHb concentration of $\leq 2.5\%$. If the MetHb concentration did not improve to within 2.5% even at a dose of 5 ppm, iNO therapy was discontinued. If no improvement was observed within 24 h after withdrawal, iNO was discontinued. If the MetHb concentration decreased by less than 2% after iNO withdrawal, iNO was reinitiated at 20 ppm and increased by 5 ppm every 5 min, continuing at a dosage that maintained the MetHb concentration at $\leq 2.5\%$.

Upon meeting the criteria for discontinuation, iNO was

Table 1. Baseline Characteristics

	iNO group (n=15)	Control group (n=15)
Sex		
Male	5 (33.3)	4 (26.7)
Female	10 (66.7)	11 (73.3)
Age (years)	62.7 $\pm$ 17.4	64.0 $\pm$ 14.4
Weight (kg)	60.3 $\pm$ 12.2	59.96 $\pm$ 13.1
BMI (kg/m ²)	22.9 $\pm$ 4.0	23.1 $\pm$ 3.3
BNP (pg/mL)	465.5 $\pm$ 400.9	285.7 $\pm$ 294.8
WHO functional class		
Class I, II	0 (0)	0 (0)
Class III	13 (86.7)	14 (93.3)
Class IV	2 (13.3)	1 (6.7)
Comorbidities		
Hypertension	4 (26.7)	5 (33.3)
Diabetes	1 (6.7)	3 (20.0)
Dyslipidemia	1 (6.7)	2 (13.3)
Cardiovascular disease	1 (6.7)	2 (13.3)
Cerebrovascular disease	0 (0)	0 (0)
Renal disease	2 (13.3)	1 (6.7)
Hepatic disease	1 (6.7)	1 (6.7)
Respiratory disease	3 (20.0)	2 (13.3)
Other notable diseases	5 (33.3)	7 (46.7)
Concomitant treatment		
ECMO	0 (0)	0 (0)
Respirator ^A	0 (0)	0 (0)
Oxygen therapy	13 (86.7)	14 (93.3)
Hemodynamics		
RAP (mmHg)	7.5 $\pm$ 4.7	5.4 $\pm$ 5.2
mPAP (mmHg)	54.5 $\pm$ 17.4	46.9 $\pm$ 12.3
PAWP (mmHg)	7.2 $\pm$ 2.96	7.7 $\pm$ 3.5
Cardiac output (L/min)	2.6 $\pm$ 0.6	3.1 $\pm$ 0.9
Cardiac index (L/min/m ²)	1.6 $\pm$ 0.3	1.9 $\pm$ 0.4
PVR (Wood units)	19.1 $\pm$ 8.6	13.0 $\pm$ 4.9
Heart rate (beats/min)	78.9 $\pm$ 13.6	78.8 $\pm$ 15.6
SvO ₂ (%)	57.3 $\pm$ 7.4	63.7 $\pm$ 10.2
SaO ₂ (%)	92.5 $\pm$ 3.8	89.7 $\pm$ 9.1
Echocardiography		
TRPG	70.9 $\pm$ 24.5	60.5 $\pm$ 16.9
IVC diameter	21.9 $\pm$ 7.2	20.7 $\pm$ 7.1
Diagnosis		
CTEPH	10	10
PAH	5	5

Categorical values are expressed as n (%) and continuous values are presented as the mean $\pm$ SD. ^ARespirator use included intubation, tracheostomy, nasal high flow, and noninvasive positive pressure ventilation. BMI, body mass index; BNP, B-type natriuretic peptide; CTEPH, chronic thromboembolic pulmonary hypertension; ECMO, extracorporeal membrane oxygenation; iNO, inhaled nitric oxide; IVC, inferior vena cava; mPAP, mean pulmonary artery pressure; PAH, pulmonary arterial hypertension; PAWP, pulmonary artery wedge pressure; PVR, pulmonary vascular resistance; RAP, right atrial pressure; SaO₂, arterial blood oxygen saturation; SvO₂, mixed venous oxygen saturation; TRPG, tricuspid regurgitation peak gradient; WHO, World Health Organization.

stopped, and participation in the clinical trial was terminated. The criteria for discontinuation were as follows: participant request to withdraw from the clinical trial after providing consent; investigator determination that the participant is unable to comply with the study protocol; death of the participant; MetHb concentration $\geq 4\%$; failure to resume administration of iNO within 24h of drug withdrawal; occurrence of an adverse event with a risk of death for which a causal relationship with the iNO cannot be excluded; and other instances in which the investigator determined that participation should be terminated.

After a 7-day treatment period, if hemodynamic stability and adequate oxygenation were maintained, the iNO dose was reduced by half every 5 min until it gradually decreased to 5 ppm. If hemodynamic stability and oxygenation were maintained at an iNO dose of 1 ppm, an attempt to stop iNO inhalation was made every 5 min. The dosing regimen, criteria for dose reduction/cessation, and discontinuation criteria for iNO therapy were based on the package insert for the drug.

Control Group The control group did not receive iNO therapy, and PVR was evaluated in the control group 30 min after patient registration and group assignment. Conventional therapy was prohibited until after PVR measurement at the 30-min time point, after which it was permitted, as in the iNO group.

Study Endpoints

The primary endpoint was the change in PVR from baseline to the 30-min time point after either iNO initiation or patient registration/group assignment (control group). Secondary endpoints included changes in hemodynamic parameters, arterial blood test findings (e.g., serum B-type natriuretic peptide [BNP] levels), echocardiographic findings (e.g., inferior vena cava [IVC] diameter), and the safety of iNO therapy. Safety was assessed based on the blood MetHb concentration, blood pressure, and adverse events during the study period.

Changes in each parameter were evaluated from screening to each measurement time point (Supplementary Table 3).

Sample Size

Given the rarity of patients with acute severe RHF due to PH, this study aimed to enroll a maximum of 30 partici-

pants based on the annual number of patients treated at the National Cerebral and Cardiovascular Center. Under a conservative scenario, we assumed a between-group mean difference of 1.2 for the primary endpoint (change in PVR) with an SD of 0.4. This effect size was selected because a prior study in Japanese patients reported a larger short-term reduction in PVR with iNO (~ 2.0 – 2.3 Wood units after ≥ 10 min of administration),¹³ whereas a small short-term change was expected in the control arm over the assessment window. The SD of 0.4 was derived from repeated pretreatment PVR measurements obtained at the National Cerebral and Cardiovascular Center, using a standardized measurement system. Under these assumptions, a total sample size of 30 patients (15 per group) would provide a power exceeding 99.9%. Even if the SD was increased to 1.2, the power remained at 75.3%. Consequently, enrollment of 30 patients in this study was considered appropriate.

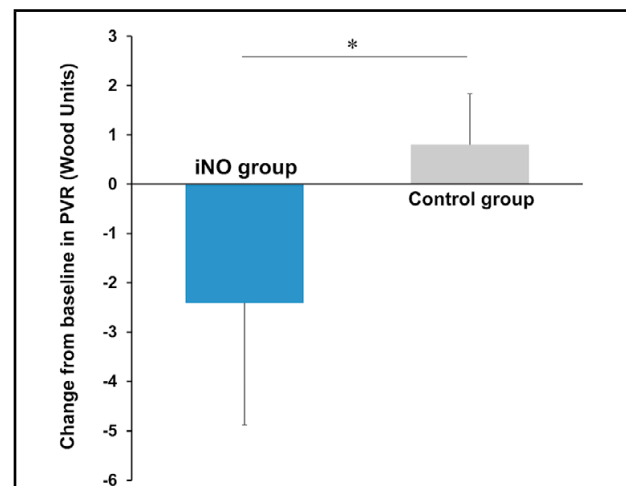


Figure 2. Change in pulmonary vascular resistance (PVR) from baseline to 30 min after inhaled nitric oxide (iNO) initiation in the iNO group and group allocation in the control group. *The estimated mean difference between the groups was -3.21 (95% confidence interval -4.633 , -1.785).

Table 2. Change in PVR From Baseline to 30 min After the Initiation of iNO Therapy (Full Analysis Set)

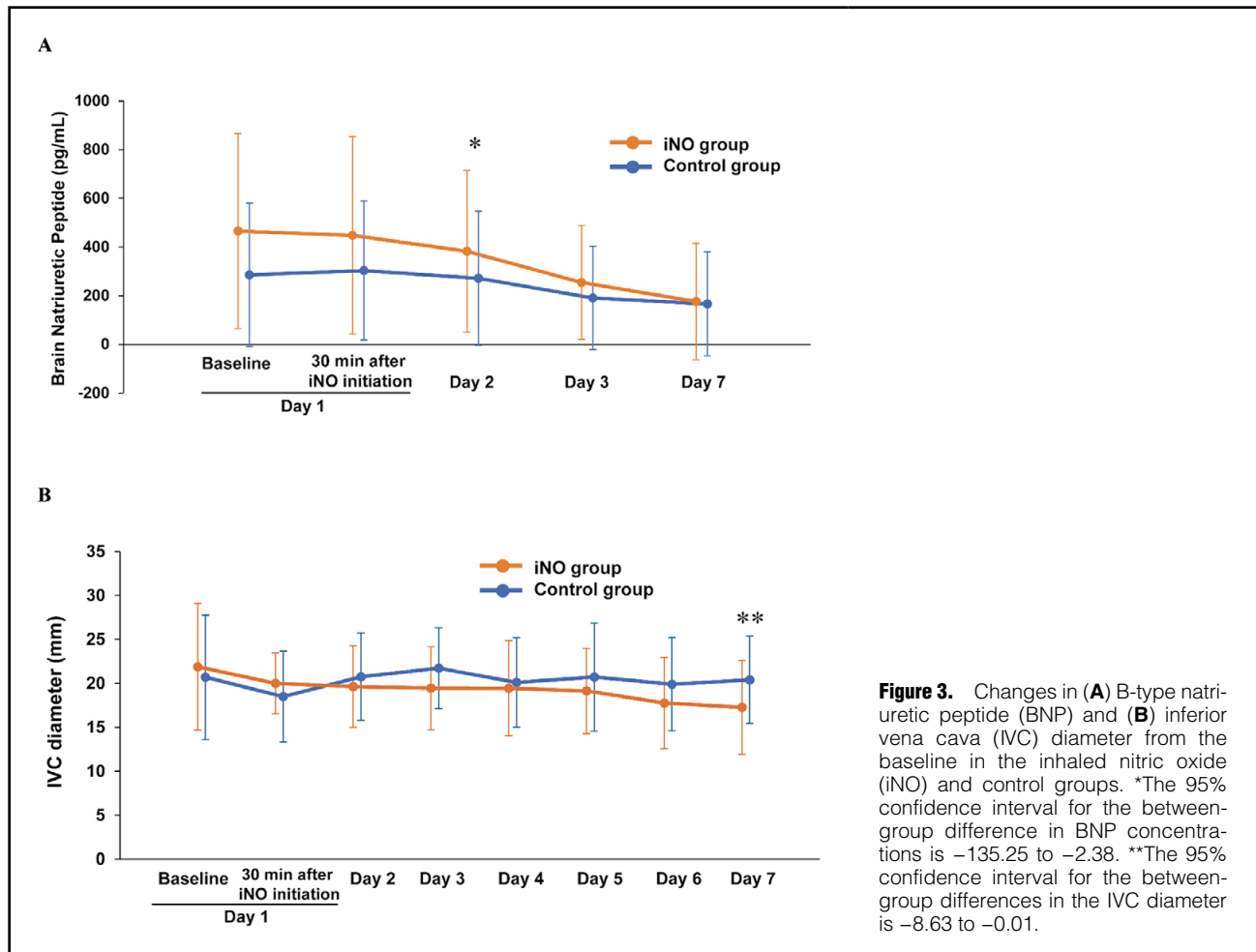
	iNO group			Control group			iNO–control between-group differences in PVR change (estimated value)
	Baseline	At 30 min	PVR change from baseline (Wood units)	Baseline	At 30 min	PVR change from baseline (Wood units)	
No. patients	15	14	14	15	15	15	
PVR (Wood units)							
Mean	19.14	17.27	–2.41	13.01	13.82	0.80	–3.21
SD	8.561	8.256	2.470	4.899	4.886	1.031	
Minimum	8.5	7.0	–6.5	5.5	6.0	–0.3	
Median	16.33	14.14	–2.67	14.01	16.05	0.50	
Maximum	36.7	35.2	1.7	19.8	20.9	2.9	
95% CI for mean			–3.835, –0.982			0.230, 1.371	–4.633, –1.785
P value			0.0029			0.0094	

Pulmonary vascular resistance (PVR) was measured at baseline (during screening on Day 1 of hospitalization) and then either 30 min after inhaled nitric oxide (iNO) initiation in the iNO group or group allocation in the control group. CI, confidence interval.

Table 3. Descriptive Statistics for Cardiac Catheterization Findings (Full Analysis Set)

	iNO group			Control group			iNO-control between-group differences (estimated value)
	Baseline	At 30 min	Change from baseline	Baseline	At 30 min	Change from baseline	
No. patients	15	14	14	15	15	15	
mPAP (mmHg)							
Mean	54.5	50.6	-4.4	46.9	48.8	1.9	-6.2
SD	17.40	17.96	3.99	12.33	12.56	1.77	
95% CI for mean	44.90, 64.17	40.27, 61.02	-6.66, -2.05	40.10, 53.76	41.84, 55.76	0.89, 2.85	-8.55, -3.90
PAWP (mmHg)							
Mean	7.2	7.3	0.4	7.7	7.1	-0.7	1.0
SD	2.96	3.24	1.39	3.47	3.65	1.05	
95% CI for mean	5.56, 8.84	5.41, 9.16	-0.45, 1.16	5.81, 9.66	5.04, 9.09	-1.25, -0.09	0.09, 1.96
RAP (mmHg)							
Mean	7.5	6.6	-1.1	5.4	5.3	-0.1	-1.1
SD	4.73	4.70	2.35	5.21	5.79	1.39	
95% CI for mean	4.91, 10.15	3.86, 9.29	-2.50, 0.21	2.52, 8.28	2.13, 8.54	-0.83, 0.70	-2.53, 0.38
Systolic PAP/systolic SBP ratio							
Mean	0.7063	0.6918	-0.0203	0.6061	0.5943	-0.0118	-0.0085
SD	0.244	0.243	0.0621	0.192	0.195	0.0403	
95% CI for mean	0.571, 0.841	0.552, 0.832	-0.0562, 0.0156	0.4999, 0.712	0.487, 0.702	-0.03641, 0.0105	-0.0048, 0.031
PVR/SVR ratio							
Mean	0.586	0.521	-0.0741	0.439	0.454	0.0151	-0.0893
SD	0.256	0.288	0.109	0.130	0.169	0.0895	
95% CI for mean	0.444, 0.728	0.35, 0.688	-0.137, -0.0011	0.367, 0.511	0.360, 0.548	-0.0345, 0.0647	-0.165, -0.00137
Cardiac output (L/min)							
Mean	2.617	2.666	0.090	3.116	3.028	-0.088	0.178
SD	0.598	0.623	0.305	0.902	0.844	0.371	
95% CI for mean	2.285, 2.948	2.306, 3.026	-0.086, 0.266	2.617, 3.616	2.561, 3.4956	-0.293, 0.117	-0.816, 0.438
Cardiac index (L/min/m²)							
Mean	1.593	1.643	0.059	1.898	1.847	-0.05	0.109
SD	0.276	0.311	0.197	0.400	0.379	0.222	
95% CI for mean	1.440, 1.746	1.464, 1.822	-0.0551, 0.172	1.676, 2.119	1.637, 2.054	-0.173, 0.072	-0.0512, 0.2692
HR (beats/min)							
Mean	78.9	77.4	-1.8	78.8	80.1	1.3	-3.1
SD	13.62	16.43	4.49	15.61	14.11	3.94	
95% CI for mean	71.33, 86.41	67.94, 86.91	-4.38, 0.81	70.16, 87.44	72.32, 87.95	-0.85, 3.52	-6.33, 0.09
SvO₂ (%)							
Mean	57.29	57.09	-0.23	63.65	61.79	-1.86	1.63
SD	7.369	6.051	4.011	10.159	9.778	8.553	
95% CI for mean	53.213, 61.374	53.599, 60.586	-2.544, 2.087	58.027, 69.279	56.379, 67.208	-6.596, 2.876	-3.522, 6.785
SaO₂ (%)							
Mean	92.54	90.77	-1.93	89.67	91.74	2.07	-4.00
SD	3.763	4.764	4.364	9.136	4.165	7.491	
95% CI for mean	90.456, 94.624	88.021, 93.522	-4.449, 0.591	84.614, 94.733	89.434, 94.046	-2.081, 6.215	-8.712, 0.721

Cardiac catheterization findings were determined at baseline (during screening on Day 1 of hospitalization) and then either 30 min after iNO initiation in the iNO group or group allocation in the control group. CI, confidence interval; HR, heart rate; cardiac index; SBP, systolic blood pressure. Other abbreviations as in Table 1.



Statistical Analysis

The primary endpoint was assessed in the FAS population by calculating the mean change in PVR and the 95% confidence interval (CI) for each treatment group. The differences between groups were summarized using the estimated value and 95% CI and compared using analysis of covariance (ANCOVA) with the value prior to the initiation of the investigational treatment as a covariate. Unless specified otherwise, hypothesis testing was conducted using a 2-tailed hypothesis with a significance level of 5%. The confidence coefficient for constructing the CIs was set at 95%.

Efficacy assessments of secondary endpoints were conducted in the FAS population. For continuous variables, the mean and 95% CI were calculated for each treatment group at each time point before and after the start of the trial treatment. The mean change from baseline and its 95% CI were calculated. For categorical variables, the results are summarized for each treatment group using the proportion of patients at each time point and the exact 95% CI. Safety assessments of the secondary endpoints were conducted in the SAS population. Primarily descriptive analyses were conducted for the number of events, cases, and proportions. Adverse events and malfunctions were tabulated according to their causality and severity. Continuous variables, such as blood MetHb concentration and systemic blood pressure, are summarized as mean $\pm$ SD or

median and range (maximum–minimum).

Results

Baseline Characteristics

In all, 32 patients provided informed consent for study participation. Two patients were deemed ineligible: one failed to satisfy selection criterion 5, and 2 violated exclusion criterion 11. Thirty patients (PAH, 10 patients; CTEPH, 20 patients) were enrolled and randomized. All patients were de novo. PAH subtypes were not confirmed during the study period. There were no AVT responders among the 10 patients with PAH. All 30 patients completed the study, with 15 receiving iNO therapy and 15 not receiving iNO (control group). iNO was administered via a face mask in all patients, and no patients used respirators. No patient discontinued iNO treatment.

In the FAS population of 30 patients, there were 15 patients each in the iNO and control groups. The baseline characteristics of all patients and other baseline values are summarized in **Table 1**. Although several hemodynamic measures appeared numerically more severe in the iNO group, the primary analysis of the change in PVR used ANCOVA adjusting for baseline PVR.

The proportion of patients receiving specific concomitant medications is presented in **Supplementary Table 4**. After

Table 4. Methemoglobin Concentrations in the Blood Over the 7-Day Study Period (Safety Analysis Set)

	iNO group							
	Day 1 screening	Day 1 (study treatment)	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
No. patients	15	15	15	15	15	15	15	15
Methemoglobin (%)								
Mean	0.43	0.51	0.98	0.89	0.81	0.86	0.86	0.81
SD	0.144	0.196	0.227	0.217	0.255	0.340	0.311	0.187
Minimum	0.2	0.2	0.5	0.3	0.3	0.3	0.1	0.5
Median	0.40	0.60	1.00	0.90	0.80	0.80	0.90	0.80
Maximum	0.7	0.8	1.5	1.3	1.3	1.5	1.5	1.2

	Control group							
	Day 1 screening	Day 1 (study treatment)	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
No. patients	15	15	15	15	15	15	15	15
Methemoglobin (%)								
Mean	0.45	0.45	0.85	0.85	0.78	0.65	0.67	0.53
SD	0.173	0.160	0.188	0.217	0.262	0.340	0.367	0.324
Minimum	0.2	0.2	0.6	0.5	0.2	0.1	0.0	0.0
Median	0.40	0.40	0.80	0.80	0.70	0.70	0.80	0.50
Maximum	0.8	0.8	1.2	1.3	1.2	1.1	1.2	1.1

iNO, inhaled nitric oxide.

assessing PVR change at 30 min in both the iNO and control groups, conventional therapy was administered, including pulmonary vasodilators, diuretics, and cardiotonic agents (dobutamine hydrochloride).

Efficacy of iNO

Primary Endpoint The change in PVR for both groups is shown in **Figure 2** and in **Table 2** for the FAS population. Neither the iNO nor control group received conventional therapy until PVR measurement at 30 min. The change in PVR at 30 min in the iNO and control groups was -2.41 ± 2.47 and 0.8 ± 1.03 Wood units, respectively. The estimated mean difference between the 2 groups was -3.21 Wood units (95% CI -4.633 , -1.785 Wood units).

ANCOVA using PVR values prior to the initiation of iNO therapy as a covariate was performed to compare the 2 groups (FAS population), revealing a significant difference in the change in PVR at 30 min between the iNO and control groups (**Supplementary Table 5**).

As a secondary analysis of the primary endpoint, ANCOVA was conducted on the change in PVR with the disease (PAH or CTEPH) as a covariate. ANCOVA of the change in PVR change with disease (PAH and CTEPH) as a covariate for between-group comparisons (FAS population) revealed significant differences in the change in PVR at 30 min between the iNO and control groups (**Supplementary Table 6**).

The change in PVR for the PPS was -2.57 ± 2.35 and -0.65 ± 0.83 Wood units in the iNO and control groups, respectively. The estimated group difference (mean) was -3.22 Wood units (95% CI -4.966 to -1.467 Wood units), which was similar to the results of the FAS analysis.

Secondary Endpoints The descriptive statistics for the RHC findings in the FAS are presented in **Table 3**. The

between-group difference in the change in mPAP from baseline screening to the 30-min time point was significant because the CI did not include zero. Descriptive statistics for arterial blood oxygen partial pressure (PaO₂), the ratio of PaO₂ to fraction of inspired oxygen (FiO₂), and the arterial blood to alveolar oxygen partial pressure ratio (AaDO₂) in the FAS are presented in **Supplementary Table 7**. A downward trend in the PaO₂/FiO₂ ratio was observed in the iNO group, whereas AaDO₂ tended to be elevated in the iNO group.

The changes in BNP and IVC diameter for the FAS are presented in **Supplementary Table 8** and **Figure 3**. Compared with the control group, the iNO group had a significantly greater decrease in BNP concentrations from Day 1 (baseline) to Day 2 (95% CI for between-group difference -135.25 , -2.38 ; **Figure 3A**). Similarly, a significant decrease in the IVC diameter was observed, with the iNO group showing a significantly greater decrease from Day 1 (baseline) to Day 7 (95% CI for between-group differences -8.63 , -0.01) than the control group (**Figure 3B**).

Subgroup Analysis For each target disease (PAH, n=10; CTEPH, n=19), the difference between the treatment groups in the primary endpoint, namely the change in PVR in the FAS, is presented in **Supplementary Table 9**. For CTEPH, the CI did not include zero, indicating a significant therapeutic effect in the iNO group.

Safety of iNO Therapy

The descriptive statistics for MetHb from the SAS are presented in **Table 4**. MetHb concentrations in neither the iNO nor control group were within a range that would raise safety concerns, and no significant difference in the MetHb concentration was observed between the 2 groups.

The occurrence of adverse events (SAS population) is presented in **Table 5**. None of the adverse events were

Table 5. Occurrence of AEs (Safety Analysis Set)

	Total (n=30)		iNO group (n=15)		Control group (n=15)	
	No. events	No. subjects (%)	No. events	No. subjects (%)	No. events	No. subjects (%)
AEs	32	19 (63.3)	17	10 (66.7)	15	9 (60)
Serious AEs	0	0 (0)	0	0 (0)	0	0 (0)
AEs by severity						
Mild	27	14 (46.7)	13	6 (40.0)	14	8 (53.3)
Moderate	5	5 (16.7)	4	4 (26.7)	1	1 (6.7)
Severe	0	0 (0)	0	0 (0)	0	0 (0)
AEs leading to death	0	0 (0)	0	0 (0)	0	0 (0)
AEs for which a causal relationship cannot be ruled out	3	3 (10)	3	3 (20)		
Serious AEs for which a causal relationship cannot be ruled out	0	0 (0)	0	0 (0)		
AEs for which a causal relationship cannot be ruled out by severity						
Mild	2	2 (6.7)	2	2 (13.3)		
Moderate	1	1 (3.3)	1	1 (6.7)		
Severe	0	0 (0)	0	0 (0)		
AEs for which a causal relationship cannot be ruled out leading to death	0	0 (0)	0	0 (0)		
Related AEs						
Investigations	2	2 (6.7)	2	2 (13.3)	0	0 (0)
Blood pressure decreased	2	2 (6.7)	2	2 (13.3)	0	0 (0)
Nervous system disorders	1	1 (3.3)	1	1 (6.7)	0	0 (0)
Headache	1	1 (3.3)	1	1 (6.7)	0	0 (0)

AE, adverse event; iNO, inhaled nitric oxide.

severe. In the iNO group, 4 (26.7%) and 13 (40.0%) patients experienced moderate and mild adverse events, respectively. Adverse events occurred in 19 of 30 (63.3%) patients, with 32 events overall (**Supplementary Table 10**): 10 of 15 (66.7%) patients with 17 events in the iNO group and 9 of 15 (60.0%) patients with 15 events in the control group. Adverse events occurring in ≥ 2 (13.3%) of the 15 individuals in the iNO group were abdominal discomfort and hypotension, each occurring in 2 (13.3%) patients (2 events). Adverse events for which causality could not be ruled out were observed in 3 patients in the iNO group: 1 (3.3%) patient with moderate hypotension and 1 (3.3%) patient with mild hypotension and headaches. The outcomes for all these adverse events were either improvement or recovery.

No deaths or serious adverse events were observed. No adverse events related to laboratory values were observed. The only adverse events for which a causal relationship with iNO therapy could not be ruled out were the 2 hypotensive events in the iNO group.

Exploratory Analysis No significant difference was observed between the iNO and control groups in the 36-Item Short Form Health Survey (SF-36).

Discussion

The PHiNO study evaluated the efficacy and safety of iNO therapy in patients with acute severe RHF due to PH. The findings of this Phase 2 investigator-initiated randomized open-label parallel-group comparison and control trial demonstrated are as follows. First, as the primary endpoint of this study, iNO significantly reduced PVR within 30 min of administration compared with PVR in the control group

measured at the same time point. Second, improvements were seen BNP levels and IVC diameter as secondary end-points, suggesting a potential reduction in right ventricular overload and congestion. Third, the safety assessments revealed no serious adverse events, with MetHb levels maintained within safe limits, confirming the tolerability of iNO in this critically ill population. Therefore, these results fill a critical knowledge gap by providing the first randomized controlled trial evidence supporting the use of iNO as a rapid-acting, selective vasodilator with potential physiological benefits in adult patients with acute RHF due to PH, as evidenced by reduced PVR and improved cardiac markers.

Rapid and selective pulmonary vasodilation by iNO¹⁹⁻²¹ addresses a critical unmet need in the management of acute severe RHF with PH, where currently approved PH drugs often have delayed effects with dose titration or systemic side effects such as hypotension. Inotropic agents carry risks such as arrhythmia, whereas diuretics carry risks such as hypotension, renal dysfunction, and electrolyte imbalance. Extracorporeal membrane oxygenation is invasive and can lead to fatal complications, such as infection and hemorrhage; therefore, it is better avoided whenever possible. In the present study, the significant decrease in PVR observed in the iNO group aligns with the pharmacological profile of the drug and its ability to swiftly reduce right ventricular afterload,¹³ thereby improving hemodynamics. The significant reduction in PVR observed within 30 min of iNO initiation in the present study suggests that iNO therapy rapidly decreases right ventricular afterload in patients with acute severe RHF due to PH. Further large-scale studies are needed to investigate the potential impact of iNO therapy

on long-term outcomes, such as mortality, in patients with acute severe RHF and PH. Although subgroup analysis revealed no significant difference in PAH, the sample size in the present study was insufficient to detect between-group differences in PAH, resulting in a wide CI. Nevertheless, the point estimate for the between-group difference was consistent with the overall population, indicating a trend towards a greater therapeutic effect in the iNO group. Furthermore, a covariance analysis, adjusting for disease type (PAH and CTEPH) as the covariate, confirmed significant treatment effects across the combined population, reinforcing the overall finding despite the underpowered subgroup analysis for PAH.

The impact of iNO therapy on PVR is significant, but its advantages for right ventricular function are also important. The secondary endpoints and exploratory analyses of this study focused on evaluating the parameters associated with right ventricular function and RHF. In the present study, there was a trend for a reduction in right atrial pressure and an increase in cardiac output, but the changes were not significant within the 30-min short treatment period; however, within 1 week of administration, the favorable results regarding BNP concentrations and IVC diameter further suggest that iNO therapy positively affects right ventricular strain and volume status, which are key determinants of prognosis in patients with RHF. Therefore, the present study suggests that iNO shows promise as a treatment option, in addition to other therapies, for patients with acute decompensated RHF due to PH.

The duration of iNO administration to facilitate the transition to approved PH medications in de novo patients is a matter for consideration. In our study, we implemented a protocol that involved continuous iNO administration for 1 week. As outlined in **Supplementary Table 4**, most patients started their existing PH medications during this 1-week protocol. Thus, we believe that maintaining iNO administration for approximately 1 week functions as a bridging therapy to approved PH medications.

The safety outcomes in this study were reassuring, with no serious adverse events reported and manageable mild-to-moderate adverse events such as hypotension and headache. Continuous MetHb monitoring and dose adjustments likely contributed to maintaining a favorable safety profile, which is essential for patients vulnerable to hemodynamic instability. The absence of systemic hypotension and selective pulmonary vasodilatory effects position iNO as a safer alternative to conventional pulmonary vasodilators, which can exacerbate circulatory compromise in acute RHF with PH. The safety of iNO may be due its immediate effectiveness, selective pulmonary vasodilation, improvement in ventilation-perfusion (V/Q) mismatch, and improvement in intrapulmonary shunting.^{12,22} iNO may be an ideal and safe acute-phase treatment for patients with RHF due to PH. In the present study, a trend towards reduced PaO₂/FiO₂ ratios and increased AaDO₂ was observed in the iNO group. Pulmonary vasodilators can cause V/Q mismatch, resulting in decreased oxygenation and AaDO₂ widening. Despite these observations, no significant reduction in PaO₂ or clinically evident respiratory failure was detected, and the use of iNO did not present any major issues. Similar occurrences may be observed with approved PH medications. However, given that iNO is an inhaled agent and is relatively less likely to induce V/Q mismatch, it may not have resulted in clinically significant hypoxia in the present study. Nonetheless, caution is advised, particularly in

patients with concurrent respiratory diseases, to ensure that it does not exacerbate hypoxemia.

iNO is currently used to treat specific conditions. First, iNO is recommended for neonatal patients with PH, such as those experiencing persistent neonatal PH or hypoxemic respiratory failure.²³ Primarily used in the neonatal intensive care unit, iNO is effective in managing postoperative PH in mechanically ventilated patients, thereby enhancing oxygenation and reducing the risk of PH crisis in neonates. Second, in adult patients undergoing cardiac surgery, iNO is frequently used to manage PH, RHF, and hypoxemia.²⁴ The study drug, INOflo, is reimbursable in numerous countries for the indication of improving hypoxic respiratory failure with PH in neonates (e.g., Japan, USA, European Union)²³ and for ameliorating PH in the perioperative period of cardiac surgery (e.g., Japan, European Union).²⁵⁻²⁷ The findings of the present study suggest a promising new therapeutic option that addresses the critical gaps in current treatment strategies for adults with acute RHF due to PH.

The limitations of the present study include its relatively small sample size, which may limit the generalizability of the findings and the ability to detect long-term outcomes or rare adverse events. The open-label design, although necessary for some practical aspects of this trial, introduced the potential for observer or patient expectation bias, which may have subtly influenced some of the assessments. This, along with the short follow-up period, restricts the assessment of sustained efficacy and safety. In addition, the study was conducted at a single center in Japan, which may limit the generalizability of the findings to other areas, healthcare systems, or diverse patient populations. Therefore, to confirm the efficacy endpoints beyond pulmonary hemodynamics, further studies with larger and more diverse patient populations and extended follow-up periods are needed to validate our results, explore long-term benefits, and determine the optimal integration of iNO within comprehensive treatment protocols for acute RHF with PH.

Conclusions

This Phase 2 investigator-initiated randomized open-label parallel-group comparison and control trial was conducted to investigate the efficacy and safety of iNO in the treatment of acute severe RHF associated with PH compared with a control group (no iNO treatment). As the primary endpoint in the present study, iNO, a rapid-acting selective pulmonary vasodilator, significantly reduced PVR from baseline to 30 min after the initiation of iNO inhalation in patients with acute severe RHF associated with PH compared with PVR in the control group determined 30 min after group allocation. As secondary endpoints, BNP levels and IVC diameter improved faster in the iNO than control group over the 7-day treatment period, indicating the benefit of early improvement in right ventricular overload and congestion. Safety assessments revealed no serious adverse events, confirming the tolerability of iNO in critically ill patients. The results of the present study suggest that iNO is a safe treatment option for adult patients with acute RHF due to PH.

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Disclosures

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IRB Information

This study was approved by the Institutional Review Board of the National Cerebral and Cardiovascular Center (Reference #1203).

Data Availability

The deidentified participant data will not be shared.

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Supplementary Files

Please find supplementary file(s);
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